Buffers

Answer key with worked solutions

A. Understanding

1. What a buffer is.

> A solution that resists changes in pH when small amounts of acid or base are added. It contains a **weak acid and its conjugate base** (or a weak base and its conjugate acid).

2. Which are buffers?

> (a) **No**: HCl is a strong acid and Cl^- is too weak a base to remove added acid.
>
> (b) **Yes**: weak acid + its conjugate base.
>
> (c) **Yes**: weak base + its conjugate acid.
>
> (d) **No**: NaOH is a strong base with no weak acid partner.

3. How the buffer works.

> (a) $CH_3COO^- + H_3O^+ \longrightarrow CH_3COOH + H_2O$
>
> (b) $CH_3COOH + OH^- \longrightarrow CH_3COO^- + H_2O$

B. Buffer pH

4. 0.10 M acid, 0.15 M acetate.

> pH = 4.74 + log(0.15 M / 0.10 M) = 4.74 + 0.18 = **4.92** (the units of M cancel inside the log)

5. 0.20 M acid, 0.10 M acetate.

> pH = 4.74 + log(0.10 M / 0.20 M) = 4.74 − 0.30 = **4.44**

6. 0.20 M NH_3, 0.10 M NH_4Cl.

> pH = 9.25 + log(0.20 M / 0.10 M) = **9.55**

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C. Adding acid

7. Add 0.020 mol HCl to 0.500 L of 0.20 M/0.20 M buffer.

> Starting moles: acid = acetate = 0.20 mol/L × 0.500 L = 0.10 mol.
>
> After reaction: acetate = 0.10 mol − 0.020 mol = 0.08 mol; acetic acid = 0.10 mol + 0.020 mol = 0.12 mol.
>
> pH = 4.74 + log(0.08 mol / 0.12 mol) = 4.74 − 0.18 = **4.56**

D. Challenge

8. Ratio for pH 5.00.

> 5.00 = 4.74 + log(ratio), so log(ratio) = 0.26 and ratio = 10^0.26 = **1.8** (about 1.8 times as much acetate as acetic acid).

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